

SE 216 – SOFTWARE PROJECT MANAGEMENT

SOFTWARE MEASUREMENTS DOCUMENT

PROJECT NAME: AI Personal Health & Fitness Coach

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Questions to identify measurements:

- Is the project staying within the API and infrastructure budget?
- How much effort went into gathering, cleaning, and labeling the health and fitness data?
- How accurate is the AI in predicting user habits and behavior patterns?
- What is the total volume of health data and scientific research successfully processed for the AI model?

Identified measurements:

- Monthly API cost / Server cost / Estimated budget
- Weekly staff-hours / Data size
- Correct predictions / Total predictions
- Number of successfully processed data rows / Number of incorrectly labeled data rows / Total MB of cleaned data

Measurement storage and collection:

What: API/Cloud infrastructure costs and staff-hours spent on project activities.

- **When:** Weekly (for team effort) and monthly (for budget billing cycles).
- **Format:** Real numbers (Currency/USD) and Integers (Hours).
- **How:** Total **active hours** are collected directly from the **Scrum board** during daily updates. The Project Manager matches these actual hours with the monthly **cloud provider invoice** and records them in the master spreadsheet.

What: Number of correct/incorrect rows, successfully processed rows, and total MB of cleaned data.

- **When:** After each data cleaning sprint or main database update.
- **Format:** Integer counts (for rows) and Real numbers (for MB).
- **How:** Extracted via **MySQL queries** and validation scripts, then entered into the project documentation by the Data Engineering team.

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What: Correct AI predictions vs. total predictions.

- **When:** After each model training iteration or automated testing session.
- **Format:** Integer counts (raw predictions) and Percentages (accuracy rate).
- **How:** Automatically calculated by the testing scripts and logged directly into the AI performance tracking dashboard.

Measurement Type	Description	Example Measurements
Budget	Compares actual project costs against the planned budget	\$250 spent / \$400 estimated in Month 2.
Team Effort	Measures team productivity by tracking the total amount of work completed through Story Points	21 Story Points completed in Sprint 1.
AI Accuracy	Evaluates if the AI model provides logical and accurate fitness and nutrition recommendations to the user	48 correct and helpful meal/workout plans / 50 total generations = 96% accuracy.
Data Quality	Measures the sensors data (GPS, steps, gyroscope) per Megabyte, factoring in the precision and variety of the tracking hardware	12,000 rows with full sensor variables (GPS + Accelerometer) / 50 MB = 240 valid rows per MB.
Data Processing Efficiency	Measures how effectively the system collects, stores, and processes large amounts of project data.	8,500 successfully processed rows/10,000 collected data rows = 85% efficiency
Defect Density	Measures the percentage of collected data that is accurate, error-free, and directly usable for analysis	9,200 accurate rows / 10,000 total collected rows = 92% usability rate.